

Implicit Attention

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Abstract

We introduce systematic joint learning of query, key and value embeddings for transformer attention via an implicit deep learning model hierarchy that refines and aligns semantics.

References

[1] 'Implicit Deep Learning'

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Armin Askari, Alicia Y. Tsai

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[2] 'Robust Implicit Networks via Non-Euclidean Contractions'

Saber Jafarpour, Alexander Davydov,
Anton V. Proskurnikov and Francesco Bullo

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[3] 'Noken Rademacher Descent'

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DOI: 10.13140/RG.2.2.20368.47367

[4] 'Attention Is All You Need'

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit,
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arXiv:1706.03762v5 [cs.CL] 6 Dec 2017

Implicit Attention

For token string $\{t_1, \dots, t_m\} \subset \mathbb{R}^r$ learn matrices

$V, K, Q, \tilde{K}$ and fixed point states $s_i \in \mathbb{R}^n, n > r$,

a latent token, we call a NoKey, so that:

$$s_i = \varphi(Vs_i + Qt_i)$$

$$\varepsilon = Ks_i + \tilde{K}t_i$$

where hyperparameter $\varepsilon \in \mathbb{R}^n$ and

$$V = \theta - \text{diag}(|\theta| \mathbb{1}_n) + \delta I_n \text{ with}$$

$$\theta \in \mathbb{R}^{n \times n}, \delta < 1, \mathbb{1}_n = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{n \times 1} \text{ and } I_n = \begin{pmatrix} 1 & 0 \\ 0 & I \end{pmatrix} \in \mathbb{R}^{n \times n}$$

ensures V is well-posed for φ

ie fixed points s_i exist and are unique.

Let query $q_i = Qt_i$, key $k_i = Ks_i$ and value $v_i = Vs_i$.
($\tilde{k}_i = \tilde{K}t_i$)

Let score be a similarity function eg a kernel.

$$\text{Define } \alpha_{ij} = \text{soft max}_{j \in [m]} \text{score}(q_i, k_j),$$

$$h_i = \sum_{j=1}^m \alpha_{ij} v_j,$$

$$\beta_{ij} = \text{soft max}_{j \in [m]} \text{score}(h_i, \tilde{k}_j),$$

and average Renyi α -entropy

$$E_{\mathcal{E}} = \frac{1}{m} \sum_{i=1}^m H_{\alpha}^i, \quad H_{\alpha}^i = H_{\alpha}(\{\beta_{ij}\}_{j=1}^m)$$

H_{α} Renyi α -entropy

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Noise Rademacher Descent (semantic refinement)

Hyperparameters: step size $\eta > 0$

of iterations N

Initial step Set $S = 0 \in \mathbb{R}^n$

Define $\Sigma_1 = (\sigma_1, \dots, \sigma_n)$ where σ_i are independent

$$\text{Prob}(\sigma_i = S + \eta) = \frac{1}{2} = \text{Prob}(\sigma_i = S - \eta)$$

Induction step Given ϵ_k , set $S = \epsilon_k$ and

draw $\epsilon' = (\sigma'_1, \dots, \sigma'_n)$ where σ'_i are independent

$$\text{Prob}(\sigma'_i = S + \eta) = \frac{1}{2} = \text{Prob}(\sigma'_i = S - \eta)$$

$$\text{Let } \epsilon_{k+1} = \begin{cases} \epsilon' & \text{if } E_{\epsilon'} < E_{\epsilon_k} \\ \epsilon_k & \text{otherwise} \end{cases}$$

Iterate N times to learn implicit model with ϵ_N :

$$s_i = \varphi(Vs_i + Qt_i)$$

$$\epsilon_N = Ks_i + \tilde{K}t_i$$

Thereby jointly learning embeddings for

queries $q_i = Qt_i$, keys $k_i = Ks_i$ and

values $v_i = Vs_i$, with lower average

Rengi α -entropy E .

Transformer attention on Q, K, V ($\mathcal{E}_N$)

Define probabilities $\alpha_{ij} = \underset{j \in [m]}{\text{softmax}} \text{score}(q_i, k_j)$

and let $j_i^\alpha = \arg \max_{j \in [m]} \alpha_{ij}$, where score

is a desired similarity function eg scaled dot-product.

Define updated query $q'_i = \text{FFN}_\Theta \left(\sum_{j=1}^m \alpha_{ij} v_j \right)$

where feed-forward net FFN with parameters Θ

is applied element-wise, to introduce non-linearity.

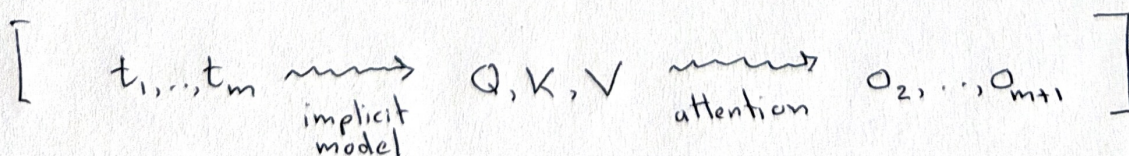
Replace q_i with q'_i and continue updating multiple times.

Output token $o_{i+1} = t_{j_i^\alpha}$

(the index right shift is for next-token inference)

From the updates, learn Θ 's to minimize

the perplexity of $o_2, \dots, o_{m+1}$



Summary

We use an implicit deep learning model [1] to jointly learn query, key, and value embeddings that refine and align semantics by lowering average Rényi α -entropy. Given these systematically found queries, keys and values, we learn feed-forward networks to minimize perplexity. This is transformer attention with Q, K, V jointly found by an implicit model of semantics; a mechanism to mitigate hallucinations.



Supplement - Cross Attention

source tokens $t'_1, \dots, t'_n$

target tokens $t_1, \dots, t_m$

Learn implicit Noken models for each language:

$$\text{For } i \in [m]: \quad s_i = \varphi(Vs_i + Qt_i)$$

$$\varepsilon = Ks_i + \tilde{K}t_i$$

$$\text{For } j \in [n]: \quad s'_j = \varphi(V's'_j + Q't'_j)$$

$$\varepsilon = K's'_j + \tilde{K}'t'_j$$

$$\text{Let } q_i = Qt_i \text{ and } k'_j = K's'_j \text{ with } v'_j = V's'_j$$

As before, define probabilities $\alpha_{ij} = \text{softmax}_{j \in [n]} \text{score}(q_i, k'_j)$

and let $j_i^\alpha = \arg \max_{j \in [n]} \alpha_{ij}$, $i \in [m]$.

Update $q'_i = \text{FFN}_\theta \left(\sum_{j=1}^n \alpha_{ij} v'_j \right)$ multiple times

Define output $o_{it_1} = t'_{j_i^\alpha}$, the translation of t_i

From the updates, learn Θ 's to minimize the perplexity of $\{o_{it_1}\}_{i=1}^m$.